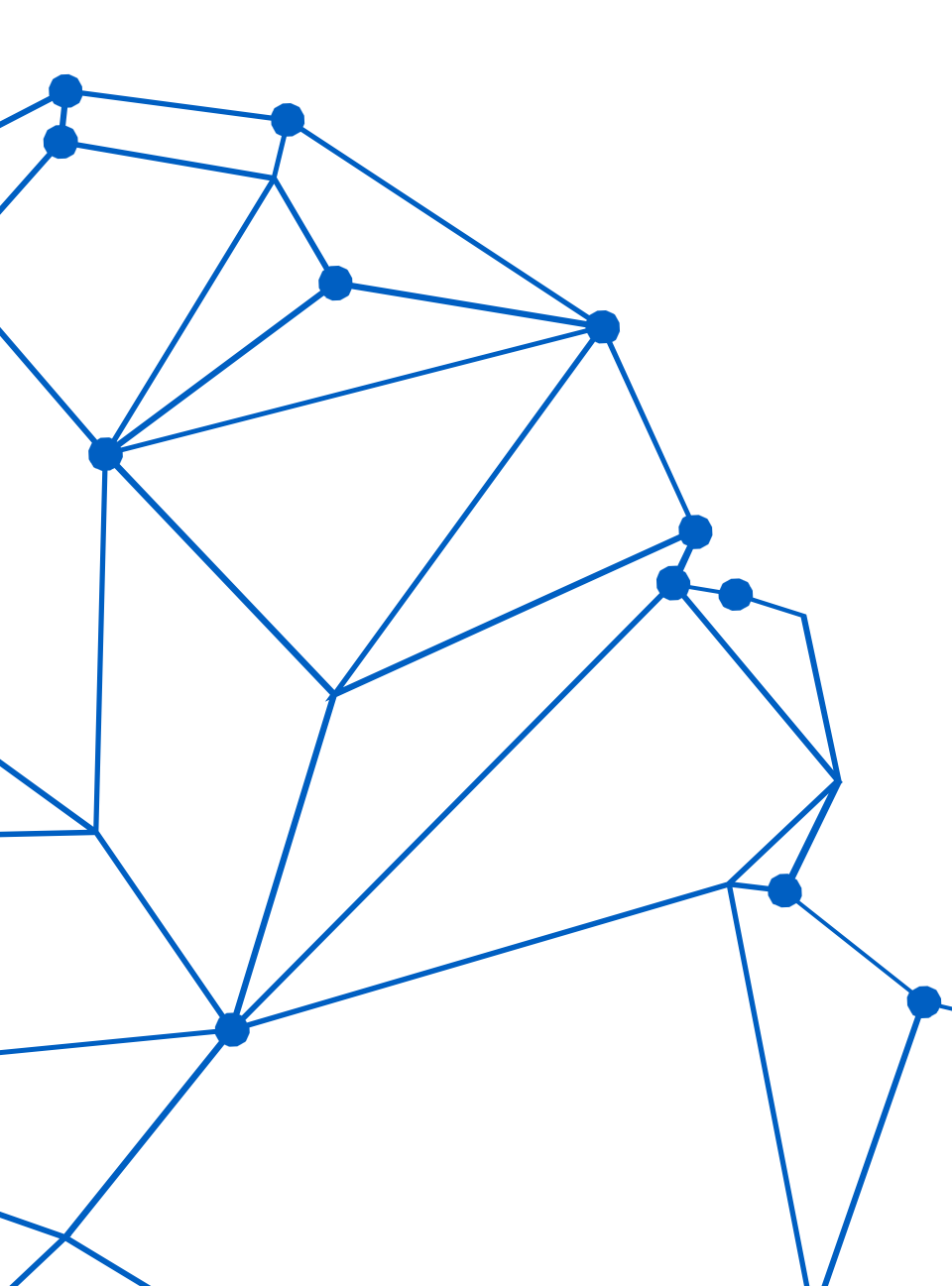
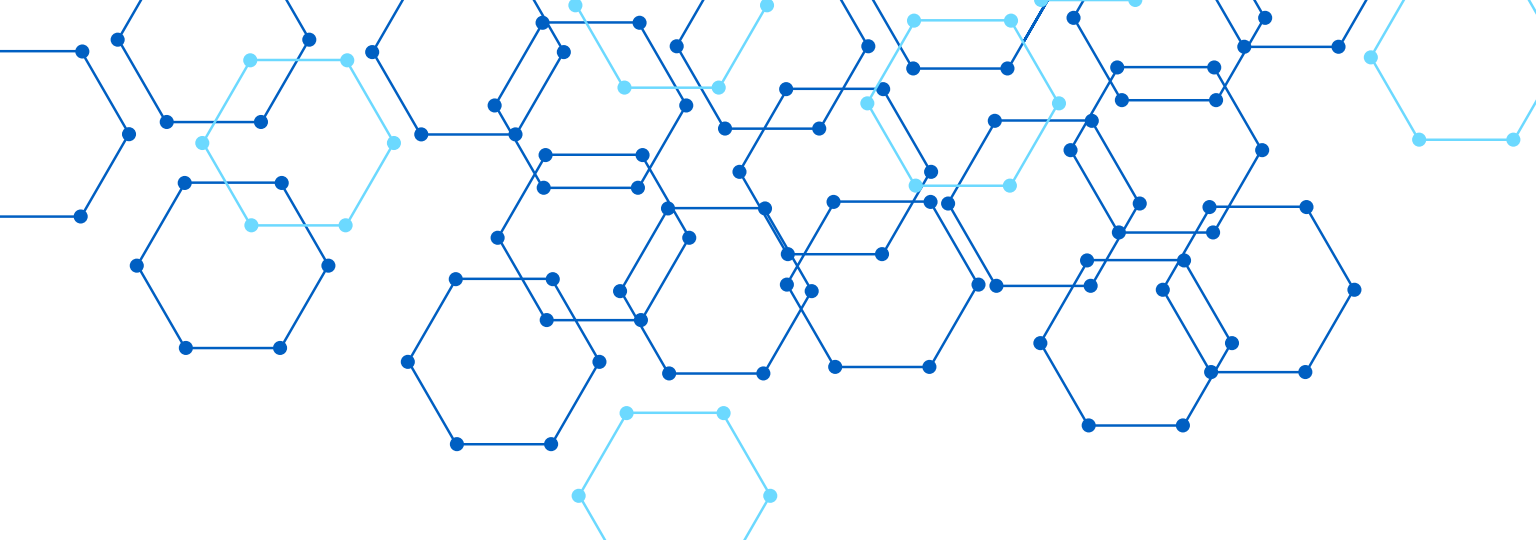
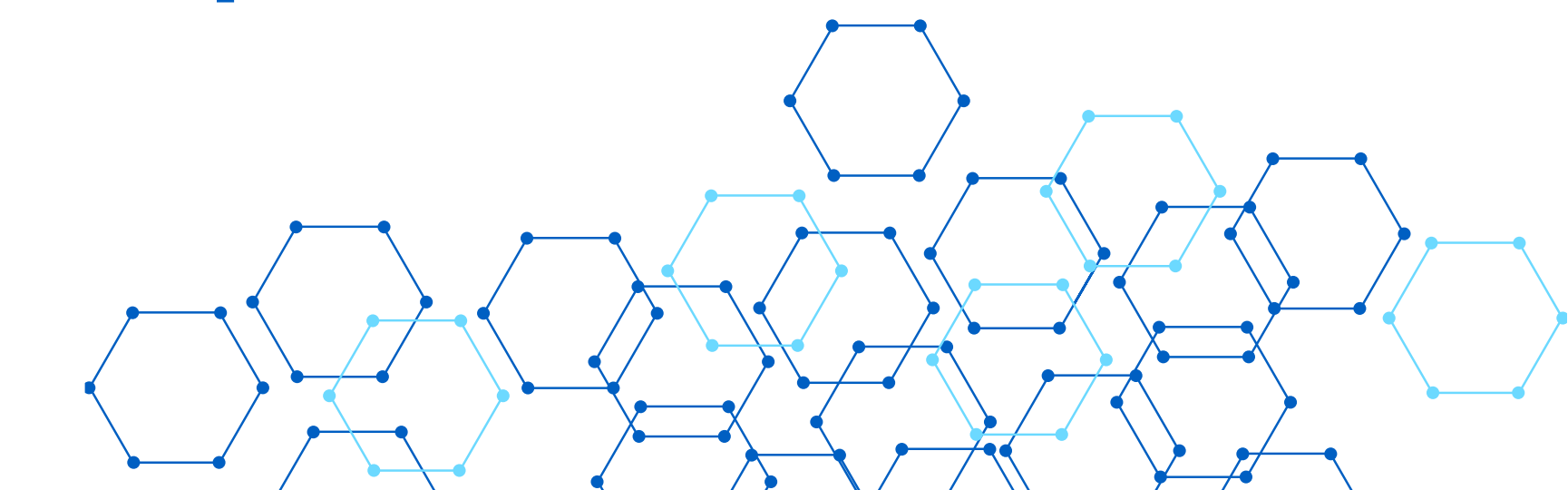




ESTIMATION FOR SOFTWARE PROJECTS

CS 191 WFQ1 - Group ACF



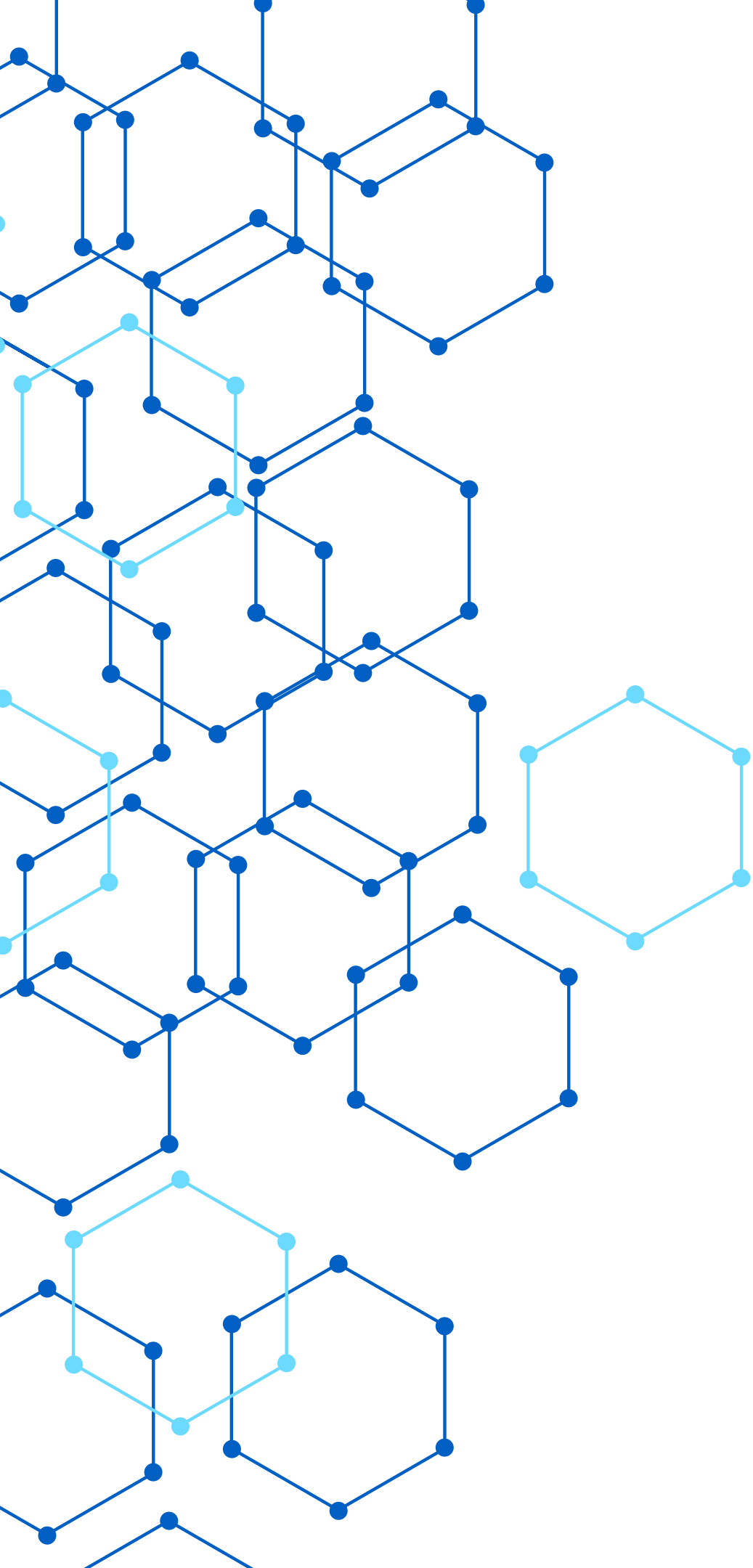
THE PLANNING DILEMMA



- Technical workers often **prefer doing technical work over planning.**
- Technical managers frequently **lack sufficient training in technical management.**



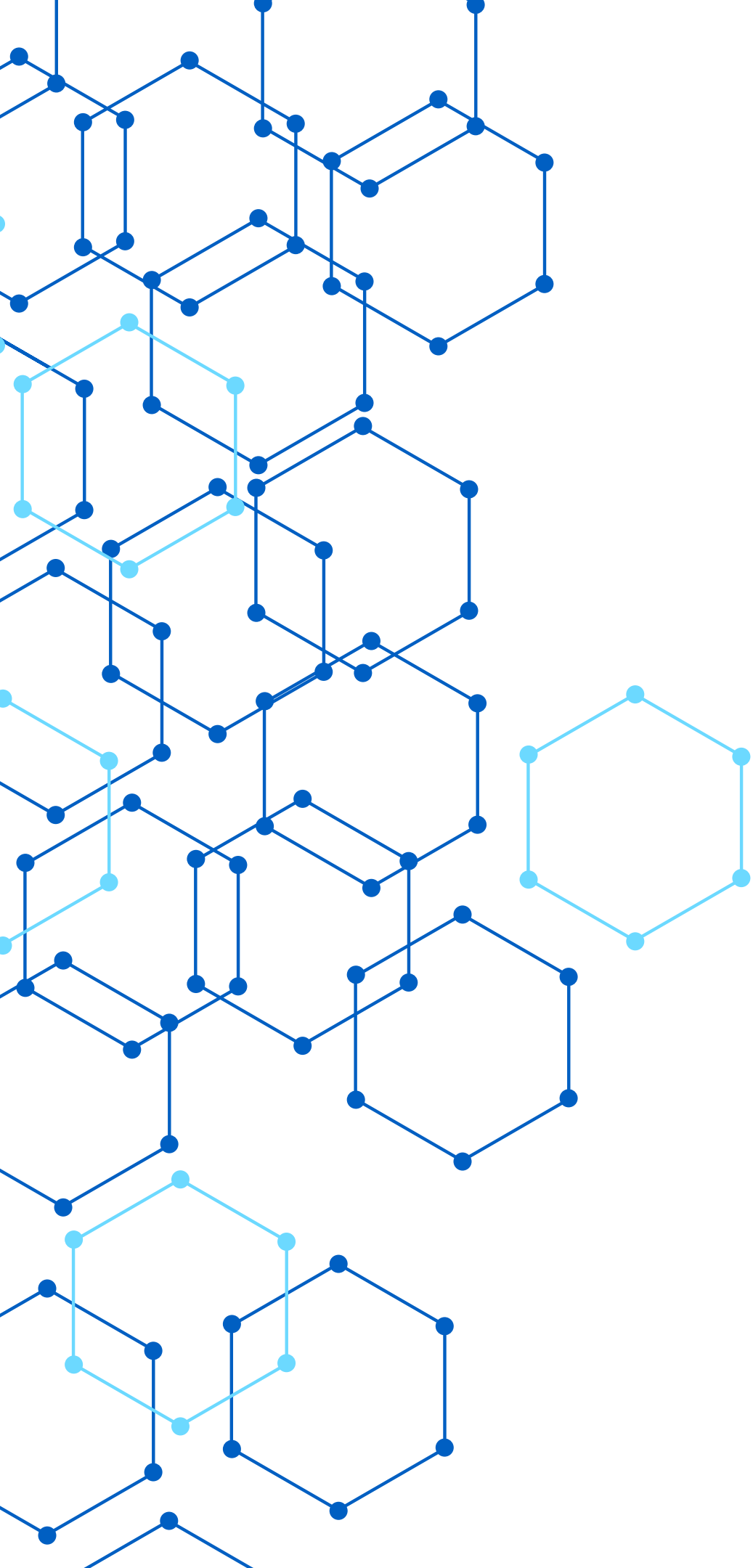
- Failure to plan is **costly**: Resolving problems upstream (early) costs far less than downstream (late). Average un-planned projects spend 80% of their time on rework.



PROJECT PLANNING

Software project management begins with **project planning**—estimating work, required resources, and elapsed time.

Scheduling establishes engineering tasks, milestones, responsibilities, and intertask dependencies.

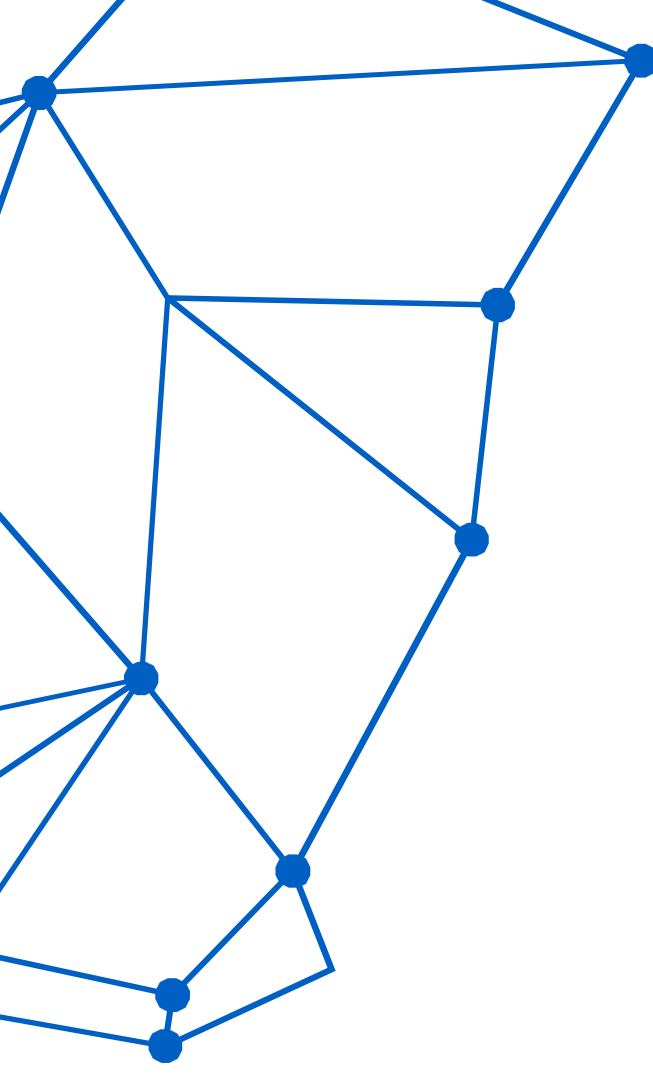


ON ESTIMATION

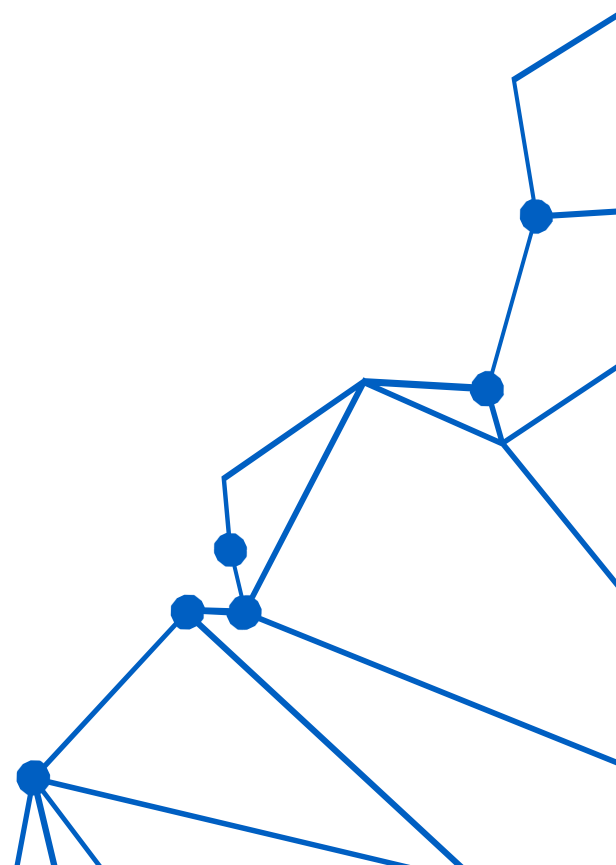
“

... our techniques of estimating are poorly developed. More seriously, they reflect an unvoiced **assumption that is quite untrue**, i.e., that **all will go well**... because we are uncertain of our estimates, software managers often lack the courteous stubbornness to make people wait for a good product.

- Frederick Brooks



KEY FACTORS AFFECTING ESTIMATION RISK

- ✓ **Project Complexity:** Relative measure based on past experience. A complex task for a beginner is routine for an experienced team.
 - ✓ **Project Size:** As size increases, interdependencies grow rapidly, making problem decomposition harder.
 - ✓ **Structural Uncertainty:** Degree to which requirements are solidified and compartmentalized.
 - ✓ **Availability of Historical Information:** Comprehensive software metrics significantly lower estimation risk.
- 

PLANNING PROCESS

Provide a framework enabling managers to make reasonable estimates of resources, cost, and schedule, defining best-case and worst-case boundaries.

**Establish
Project Scope**

**Determine
Feasibility**

**Analyze
Risks**

**Define
Resources**

**Estimate Cost
and Effort**

**Develop a
Schedule**



Establish
Project Scope

SOFTWARE SCOPE

Bounds the system by describing functions, features, inputs/outputs, performance, constraints, and reliability.

Defined using *one of two techniques*:

- **Narrative descriptions** (from stakeholder communication) or
 - **Use Cases** (developed by end users).
- 

PLANNING PROCESS

Provide a framework enabling managers to make reasonable estimates of resources, cost, and schedule, defining best-case and worst-case boundaries.

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**Estimate Cost
and Effort**

**Develop a
Schedule**

Determine Feasibility

Technology:

Is a project technically *feasible*?

Is it within the state of the art?

Can defects be reduced to a level matching the application's needs?

Finance:

Is it financially *feasible*?

Can development be completed at a cost the software organization, its client, or the market can afford?

Time:

Will the project's *time-to-market* beat the competition?

Resources:

Does the organization *have the resources* needed to succeed?

- Putnam & Myers

PLANNING PROCESS

Provide a framework enabling managers to make reasonable estimates of resources, cost, and schedule, defining best-case and worst-case boundaries.

**Establish
Project Scope**

**Determine
Feasibility**

**Analyze
Risks**

**Define
Resources**

**Estimate Cost
and Effort**

**Develop a
Schedule**

PLANNING PROCESS

**Analyze
Risks**

will be discussed by **Chapter 28**

**Develop a
Schedule**

will be discussed by **Chapter 27**

PLANNING PROCESS

Provide a framework enabling managers to make reasonable estimates of resources, cost, and schedule, defining best-case and worst-case boundaries.

**Establish
Project Scope**

**Determine
Feasibility**

**Analyze
Risks**

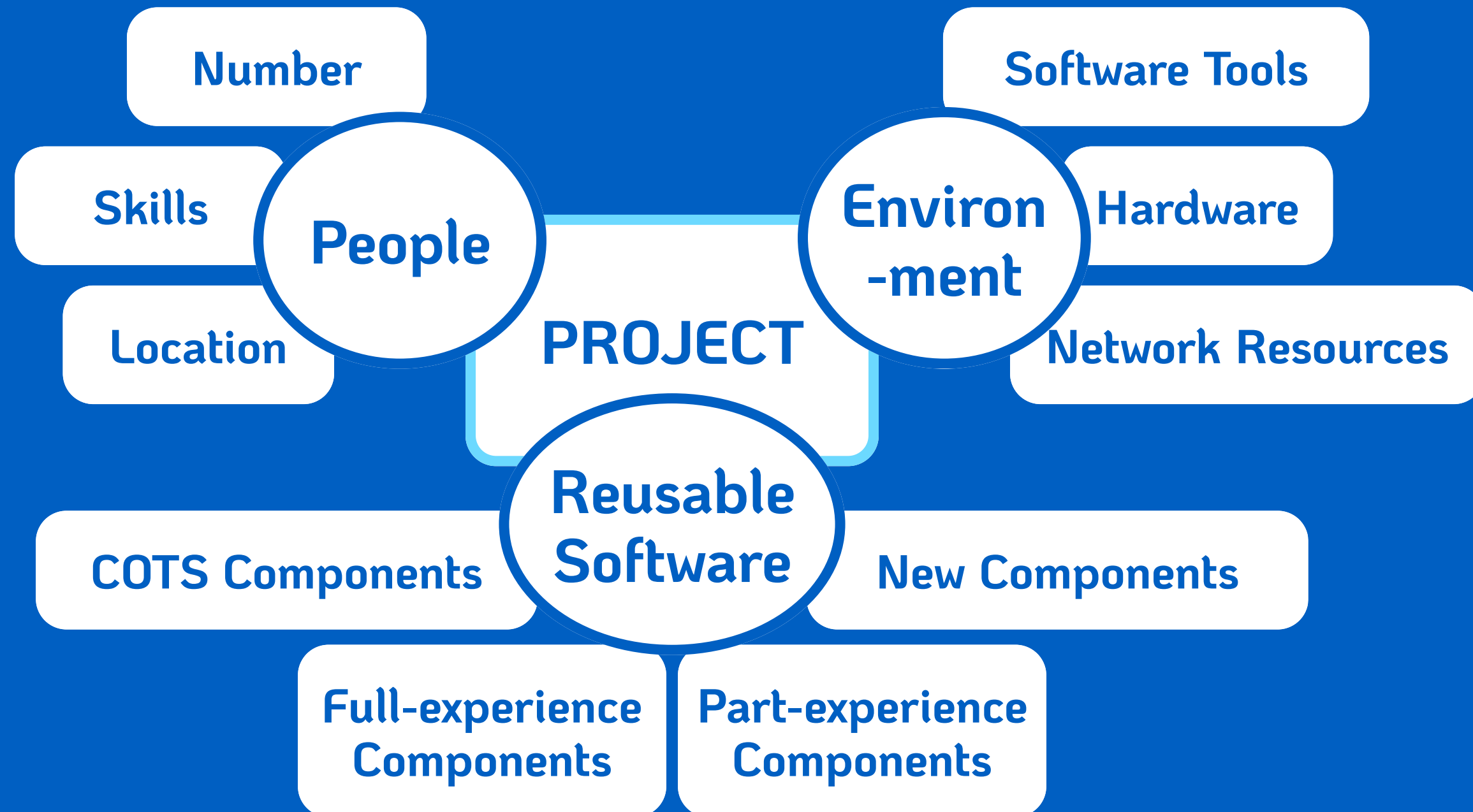
**Define
Resources**

**Estimate Cost
and Effort**

**Develop a
Schedule**

**Define
Resources**

RESOURCES



PLANNING PROCESS

Provide a framework enabling managers to make reasonable estimates of resources, cost, and schedule, defining best-case and worst-case boundaries.

**Establish
Project Scope**

**Determine
Feasibility**

**Analyze
Risks**

**Define
Resources**

**Estimate Cost
and Effort**

**Develop a
Schedule**

Estimate Cost and Effort



26.5 SOFTWARE PROJECT ESTIMATION



Will never be an exact science



Too many variables can affect the ultimate cost of software and effort applied to develop it (e.g., human, technical, environmental, political, etc.).



however, we can follow/develop a series of systematic steps to produce an estimate with acceptable risks

26.5 APPROACHES TO SOFTWARE ESTIMATION



Not Practical

- Delay estimation until late in the project
- Base estimates on similar projects that have already been completed.



Practical

- Use relatively simple **decomposition techniques** to generate project cost and effort estimates
- Use one or more **empirical models** for software cost and effort estimation.

26.6 DECOMPOSITION TECHNIQUES

Decomposition techniques take a **divide-and-conquer** approach to software project estimation. By decomposing a project into **major functions** and **related software engineering activities**, cost and effort estimation can be performed in a stepwise fashion.

26.6.1 SOFTWARE SIZING



Before an estimate can be made, you must understand the scope of the software to be built and generate an estimate of its “size.”



Project estimate is only as good as the estimate of the size of the work to be accomplished, software sizing represents your first major challenge as a planner.



In the context of project planning, size refers to a quantifiable outcome of the software project.

26.6.1 SOFTWARE SIZING



Approaches to Measuring Software Size

- Direct Approach: Lines of Code (LOC)
- Indirect Approach: Function Points (FP)



Key Factors for Estimating Size

- Project type and its application domain
- Total functionality delivered (measured in FPs)
- Number of components to be delivered
- Extent of modifications required for existing components

ESTIMATION OF SOFTWARE SIZE



Software sizing is full of uncertainty



Calculate an expected value S using optimistic (S_{opt}), most likely (S_m), and pessimistic (S_{pess}) estimates.

- small probability of the actual size will fall outside of the optimistic and pessimistic values

$$S = \frac{S_{opt} + 4S_m + S_{pess}}{6}$$

EXAMPLE OF LOC-BASED ESTIMATION

Function	Estimated LOC
User interface and control facilities (UICF)	2,300
Two-dimensional geometric analysis (2DGA)	5,300
Three-dimensional geometric analysis (3DGA)	6,800
Database management (DBM)	3,350
Computer graphics display facilities (CGDF)	4,950
Peripheral control function (PCF)	2,100
Design analysis modules (DAM)	8,400
<i>Estimated lines of code</i>	<i>33,200</i>

EXAMPLE OF LOC-BASED ESTIMATION

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Design analysis modules (DAM)	8,400
<i>Estimated lines of code</i>	<i>33,200</i>

LOC estimates for 3d geometric analysis

- $S_{\text{opt}} = 4600 \text{ LOC}$, $S_{\text{m}} = 6900 \text{ LOC}$, $S_{\text{pess}} = 8600 \text{ LOC}$

$$S = \frac{4600 + (4)(6900) + 8600}{6} = 6800$$

EXAMPLE OF LOC-BASED ESTIMATION

Organizational Average Productivity	=620 LOC/pm
Burdened Labor Rate	=\$ 8000
cost per line of code	=8000 / 620 = \$13
total estimated project cost	= 33,200 x \$13 = \$ 431,600

EXAMPLE OF FP-BASED ESTIMATION

Factor	Value
Backup and recovery	4
Data communications	2
Distributed processing	0
Performance critical	4
Existing operating environment	3
Online data entry	4
Input transaction over multiple screens	5
Master files updated online	3
Information domain values complex	5
Internal processing complex	5
Code designed for reuse	4
Conversion/installation in design	3
Multiple installations	5
Application designed for change	5
Value adjustment factor	1.17

EXAMPLE OF FP-BASED ESTIMATION

Information domain value	Opt.	Likely	Pess.	Est. count	Weight	FP count
Number of external inputs	20	24	30	24	4	97
Number of external outputs	12	15	22	16	5	78
Number of external inquiries	16	22	28	22	5	88
Number of internal logical files	4	4	5	4	10	42
Number of external interface files	2	2	3	2	7	15
Count total						320

EXAMPLE OF FP-BASED ESTIMATION

Value adjustment factor	$= 0.65 + 0.01 \times (\text{total Factor value})$ $= 0.65 + 0.01 \times 52 = 1.17$
FP Estimated	$= \text{count total} \times \text{value adjustment factor}$ $= 320 \times 1.17 = 375$
organizational average productivity	$= 6.5 \text{ FP/pm}$
Burdened Labor Rate	$= \$ 8000$
cost per FP	$= 8000 / 6.5 = \$ 1230$
total estimated project cost	$= 375 \times 1230$ $= \\$ 461,250$

PROCESS BASED ESTIMATIONS



Process based Estimations

- Estimate is based on the process that will be used
- Decomposes the project into framework activities
- estimate the effort (e.g., **person-months**) that will be required to accomplish each software process activity

RECONCILING ESTIMATES



- Merge results from different estimation techniques into a single estimate for effort, duration, or cost.
- Significant discrepancies between the estimates requires reevaluation of information used to make the estimates

26.7 EMPIRICAL ESTIMATION MODELS

- Model based on experience (historical data)
- Empirical models use empirically derived formulas to predict effort as a function of LOC or FP.

STRUCTURE OF ESTIMATION MODELS

A typical estimation model is derived using regression analysis on data collected from past software projects. The overall structure of such models takes the form

$$E = A + B \times (e_v)^C$$

- A, B, and C are empirically derived constants,
- E is effort in person-months, and
- e_v is the estimation variable (either LOC or FP)

STRUCTURE OF ESTIMATION MODELS

Examples:

- **LOC-Oriented Estimation Models**

$$E = 5.2 \times (\text{KLOC})^{0.91}$$

Walston-Felix model

$$E = 5.5 + 0.73 \times (\text{KLOC})^{1.16}$$

Bailey-Basili model

$$E = 3.2 \times (\text{KLOC})^{1.05}$$

Boehm simple model

$$E = 5.288 \times (\text{KLOC})^{1.047}$$

Doty model for $\text{KLOC} > 9$

- **FP-Oriented Estimation Models**

$$E = -91.4 + 0.355 \text{ FP}$$

Albrecht and Gaffney model

$$E = -37 + 0.96 \text{ FP}$$

Kemerer model

$$E = -12.88 + 0.405 \text{ FP}$$

Small project regression model

THE SOFTWARE EQUATION

- a model that assumes a specific distribution of effort over the life of a software development project
- derived from productivity data collected for over 4,000 contemporary software projects

$$E = \frac{LOC \times B^{0.333}}{P^3} \times \frac{1}{t^4}$$

THE SOFTWARE EQUATION

$$E = \frac{LOC \times B^{0.333}}{P^3} \times \frac{1}{t^4}$$

- E = effort in person-months or person-years
- t = project duration in months or years
- B = 'special skills factor'
- P = 'productivity parameter'

26.8 ESTIMATION FOR OBJECT ORIENTED PROJECTS

- ✓ Develop conventional estimates
- ✓ Develop use cases and determine use-case count
- ✓ Determine key classes
- ✓ Categorize interface type and determine multiplier
- ✓ Estimate Work
- ✓ Cross-check the class-based estimate



26.8 ESTIMATION FOR OBJECT ORIENTED PROJECTS

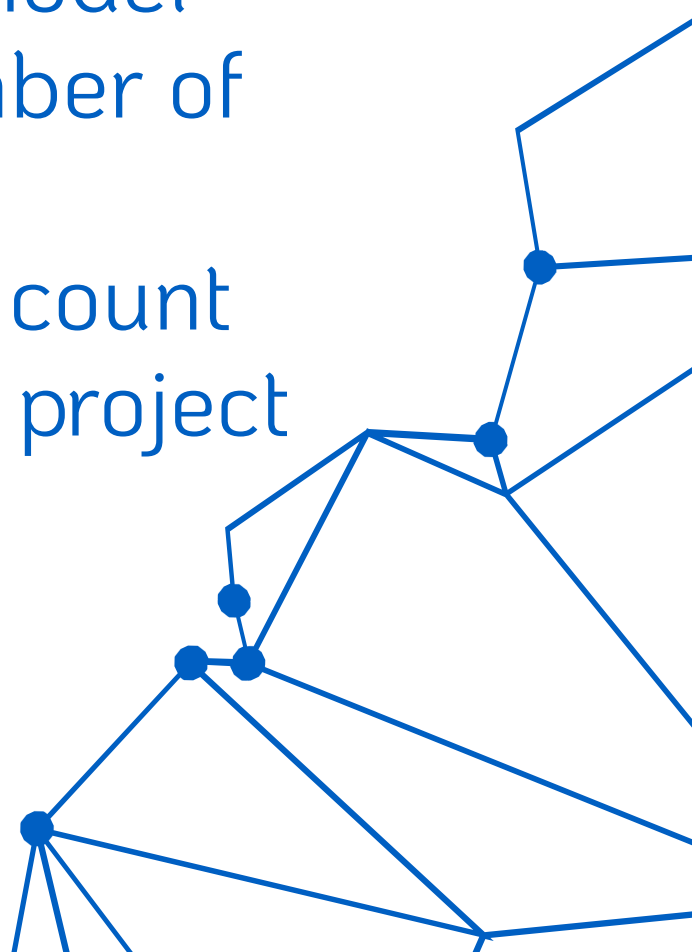


Develop conventional estimates

- Effort decomposition
- Function Point (FP) analysis
- Other applicable conventional methods



Develop use cases and determine use-case count

- Develop use cases from the requirements model
 - Determine the number of use cases
 - Recognize that the count may change as the project progresses
- 



26.8 ESTIMATION FOR OBJECT ORIENTED PROJECTS

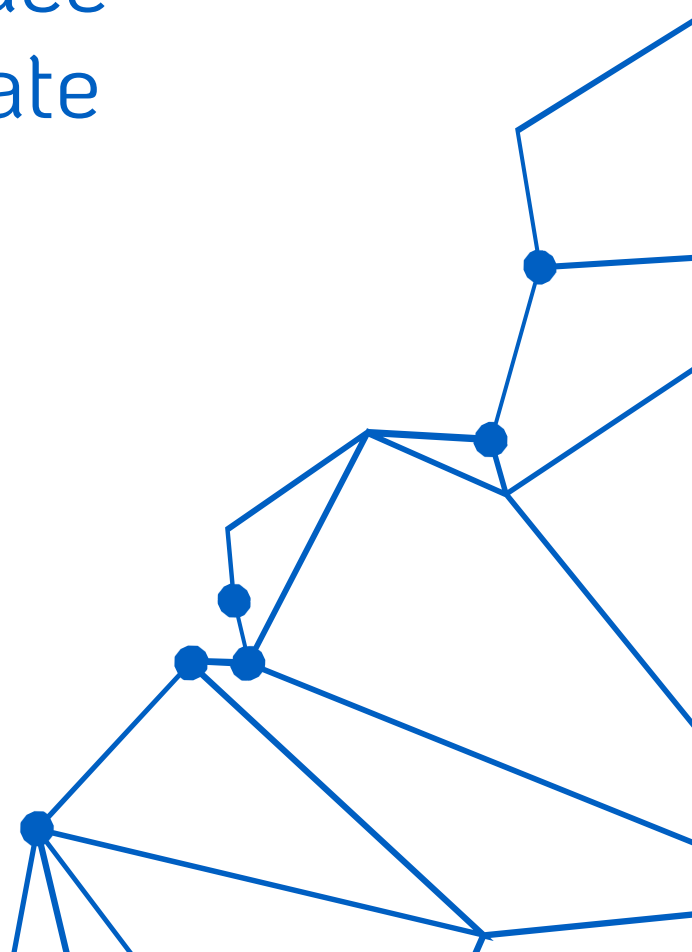


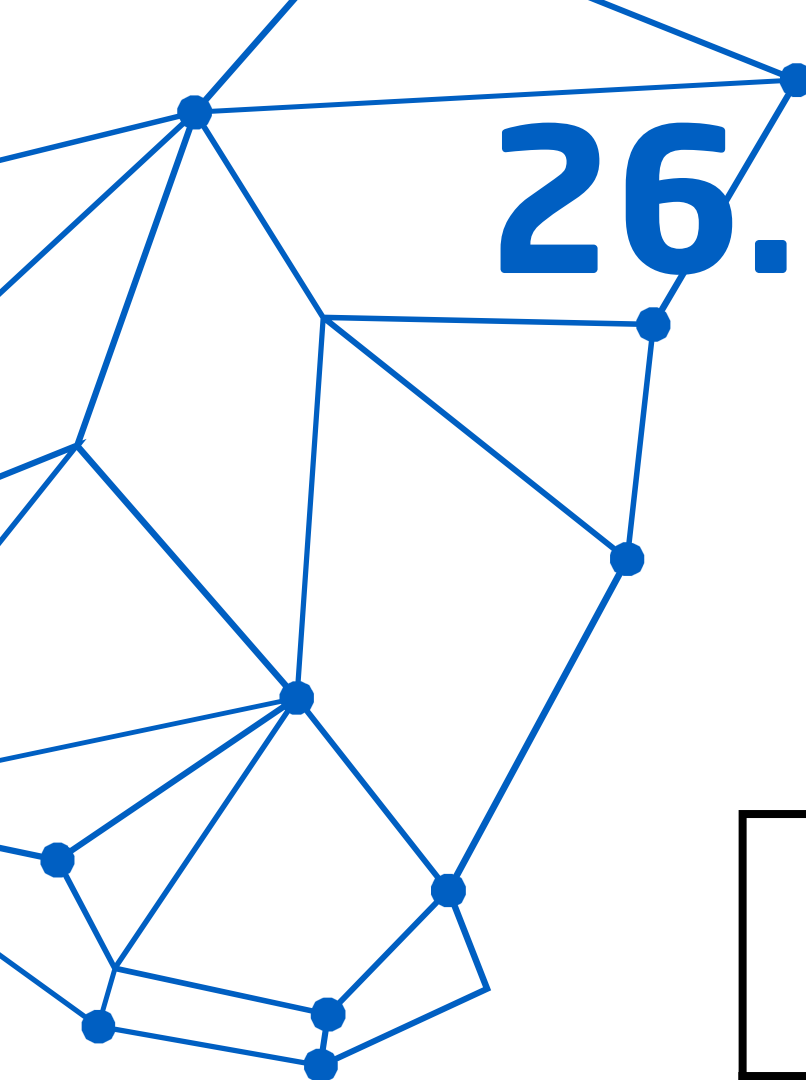
Determine key classes

- Identify the number of key classes
- Also called analysis classes



Categorize interface type and determine multiplier

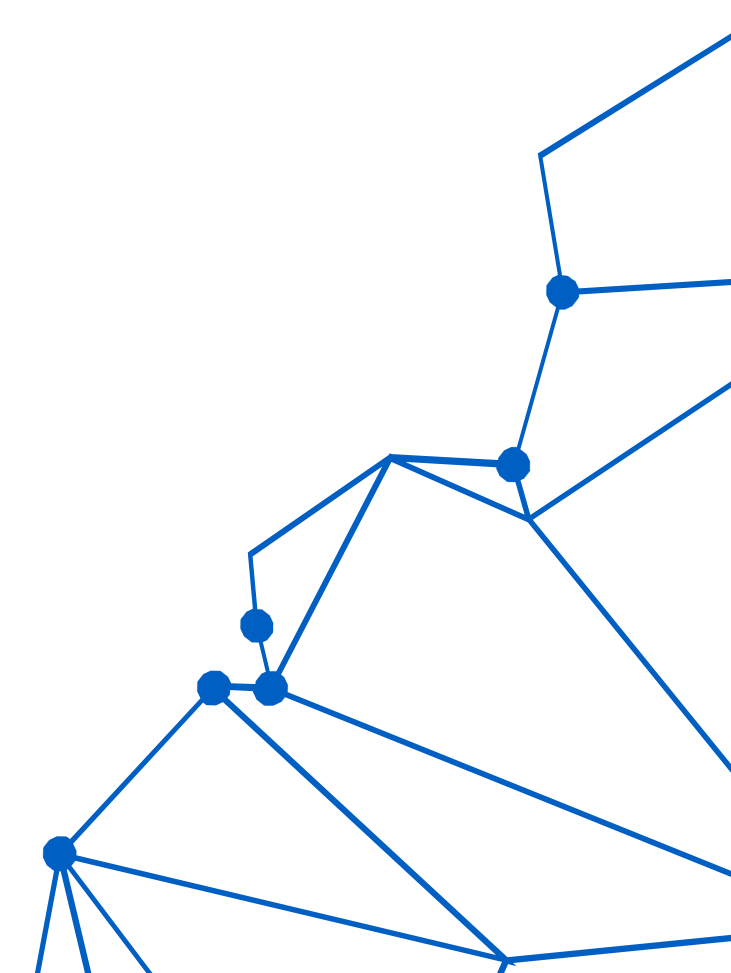
- Categorize the application's interface
 - Apply the appropriate multiplier
- 



26.8 ESTIMATION FOR OBJECT ORIENTED PROJECTS

Determining the Multiplier

Interface Type	Multiplier
No GUI	2.0
Text-based UI	2.25
GUI	2.5
Complex GUI	3.0





26.8 ESTIMATION FOR OBJECT ORIENTED PROJECTS

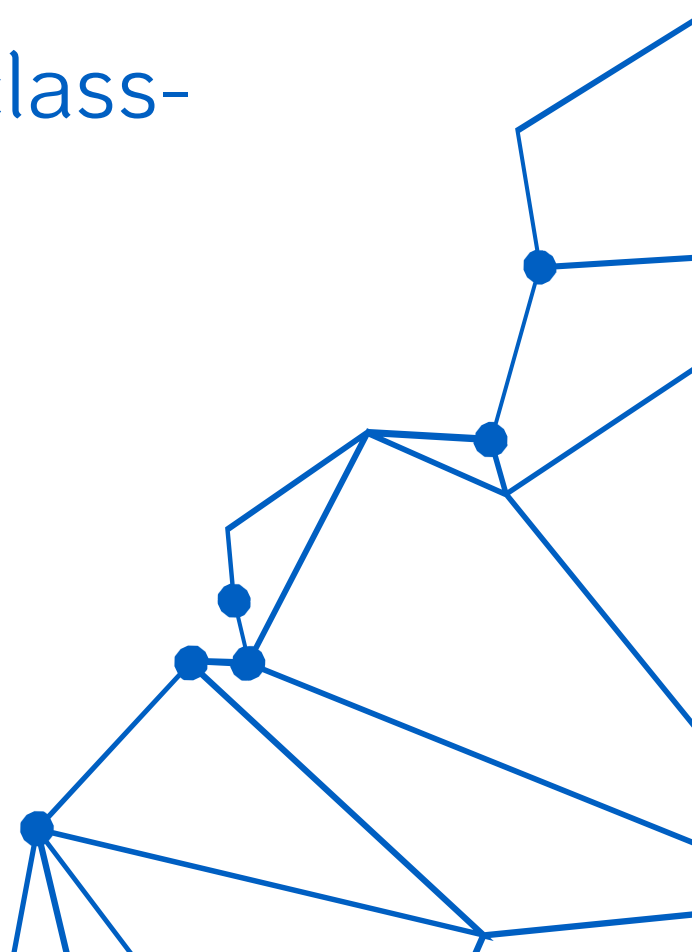


Estimate Work

- Total classes = key classes + support classes
- Multiply total classes by average work units per class
- Suggested average: 15–20 person-days per class



Cross-check the class-based estimate

- Multiply average work units per use case
 - Compare with the class-based estimate
- 

26.9 SPECIALIZED ESTIMATION TECHNIQUES

- Conventional estimation techniques can be applied to most projects
- Some projects require specialized approaches
- Particularly relevant when:
 - Project duration is very short
 - Requirements are continuously changing
- Two specialized techniques:
 - 26.9.1 Estimation for Agile Development
 - 26.9.2 Estimation for WebApp Projects

```
_form" >  
>Authentication Failed</b></form  
="dError1">Please contact the admin  
  
tus>-1</saml-auth-status>  
  
w.top.location='/php/login.php'
```

26.9.1 ESTIMATION FOR AGILE DEVELOPMENT

- Q Consider each user scenario separately
- Q Decompose the scenario into software engineering tasks
- Q Estimate effort or scenario volume
- Q Sum estimates for the scenario
- Q Sum all scenario estimates for the increment

TWO WAYS TO ESTIMATE



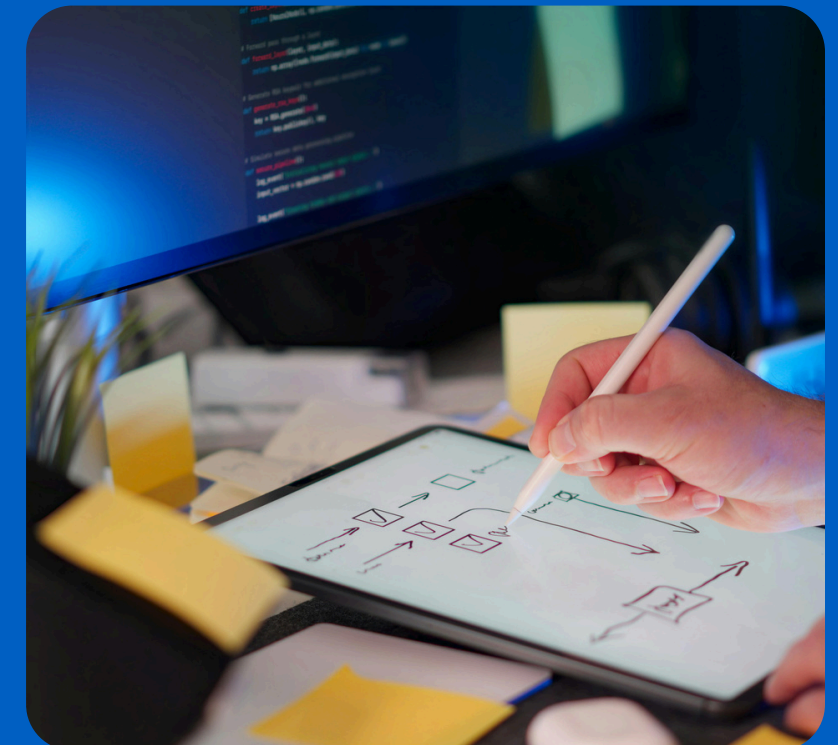
Effort-oriented

- Estimate the effort required for each task
- Based on:
 - Historical data
 - Empirical models
 - Experience



Volume-oriented

- Estimate the "volume" of the scenario
- LOC
- Function Points
- Use-case count



PURPOSE OF AGILE ESTIMATION

Even though increments are short—typically 3–6 weeks—estimation remains important.

It helps to:

- Ensure the number of scenarios fits available resources
- Establish a basis for allocating effort
- Support planning of the increment

Short duration $\neq$ no planning

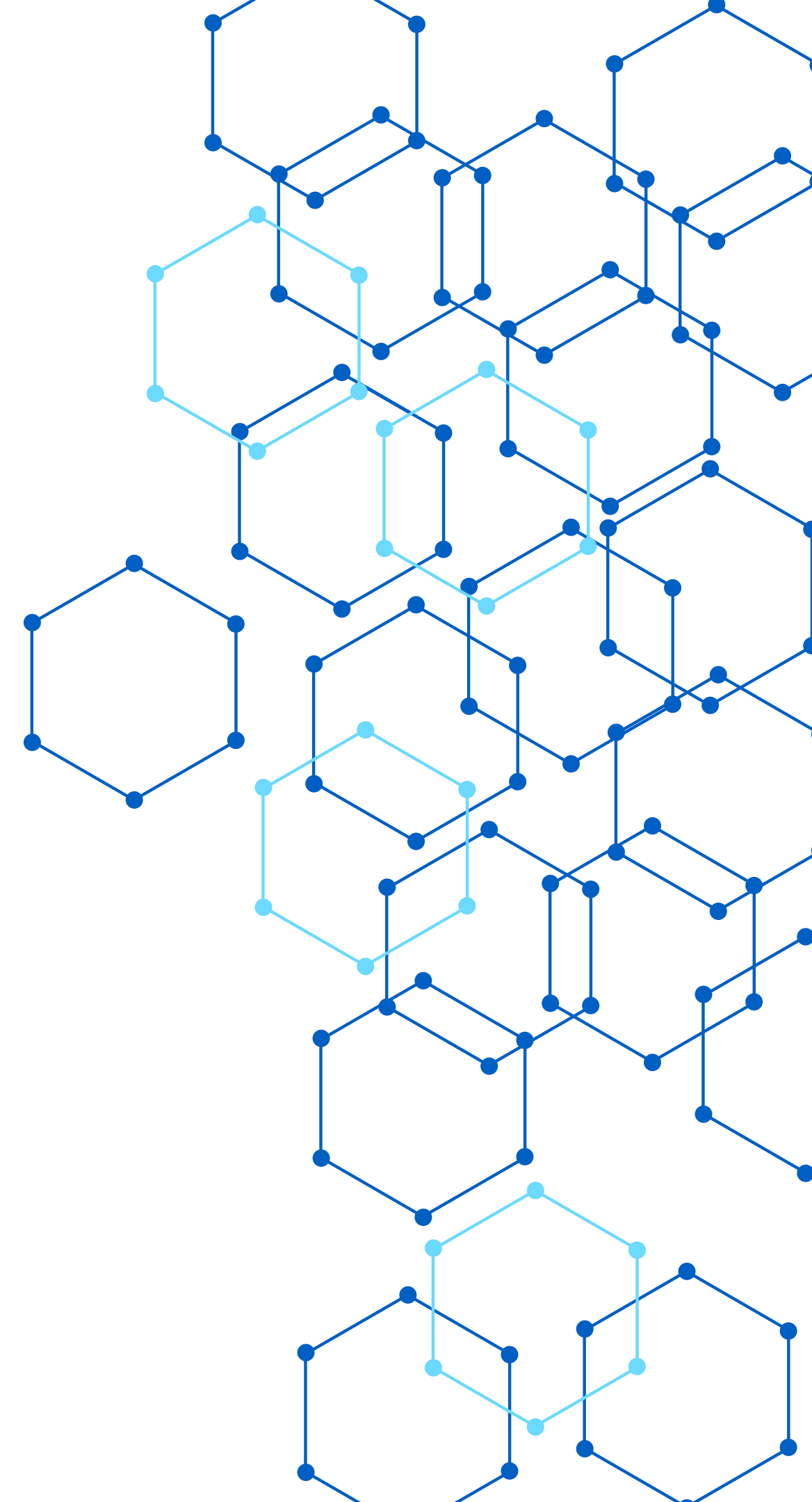
26.9.2 ESTIMATION FOR WEBAPP PROJECTS

WebApp projects often use an agile process model.

A modified Function Point approach can be used.

Function Point Elements

- Inputs
- Outputs
- Tables
- Interfaces
- Queries



26.9.2 ESTIMATION FOR WEBAPP PROJECTS

Element	WebApp Interpretation
Inputs	Input screens/forms, maintenance screens
Outputs	Static/dynamic pages and reports
Tables	Logical database tables / XML objects
Interfaces	Logical files crossing system boundaries
Queries	Externally published/message-oriented interfaces



OTHER WEBAPP PREDICTOR VARIABLES

✓ Application

- Page count
- Media count
- Function count

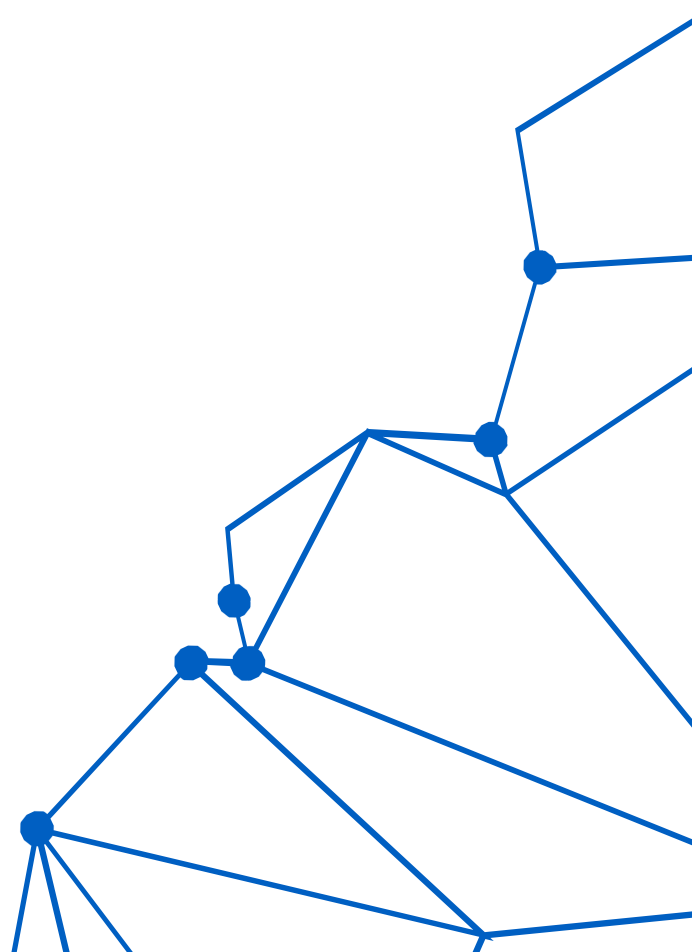
✓ Media Characteristics

- Media duration

✓ Web Page Characteristics

- Page complexity
- Linking complexity
- Graphic complexity

✓ Functional Characteristics

- Code length
 - Reused code length
- 



26.10 THE MAKE/BUY DECISION

Software can be acquired instead of developed internally.



Acquisition Points

- Buy off-the-shelf software
- Reuse existing software and modify it
- Build custom software internally
- Contract an outside vendor



Key Questions

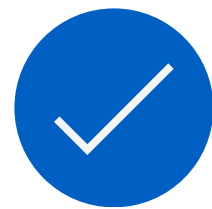
- Which option meets requirements?
 - Which option meets the delivery schedule?
 - Which option provides the best overall cost?
- 

THINGS TO CONSIDER



Delivery

- Will the acquired software be delivered sooner?



Customization

- Is acquisition + customization cheaper than developing internally?



Support

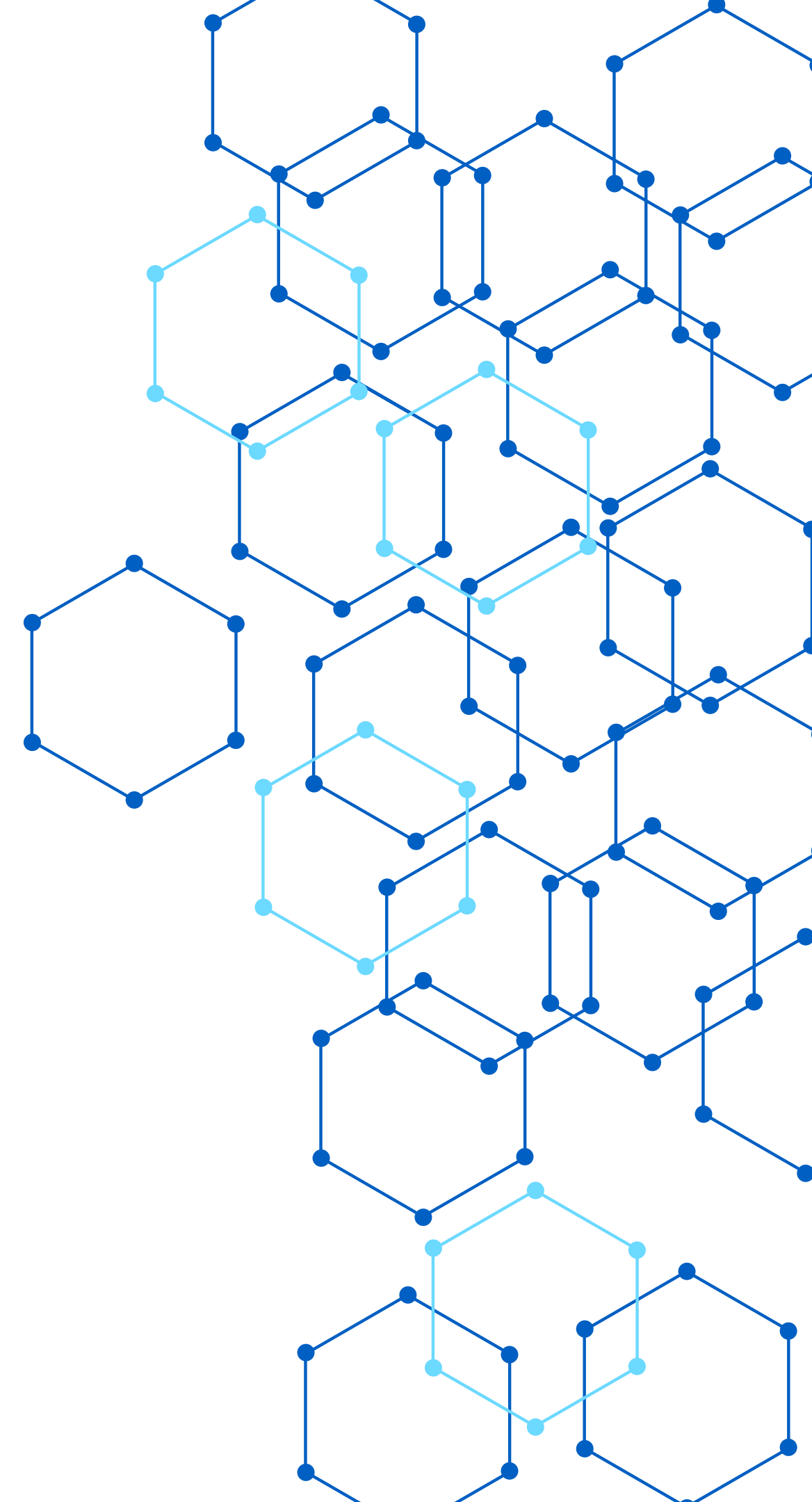
- Is outside support cheaper than internal support?

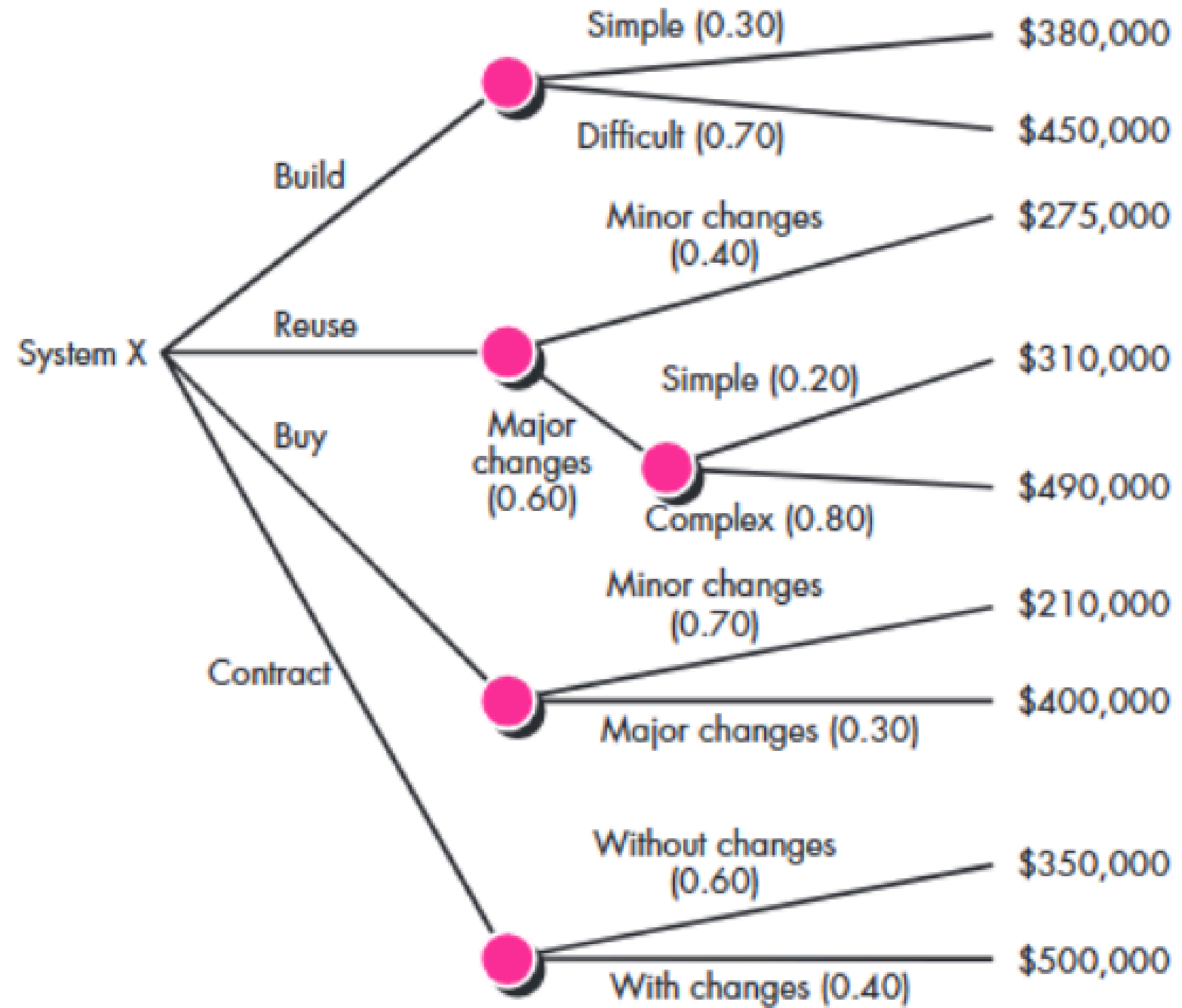
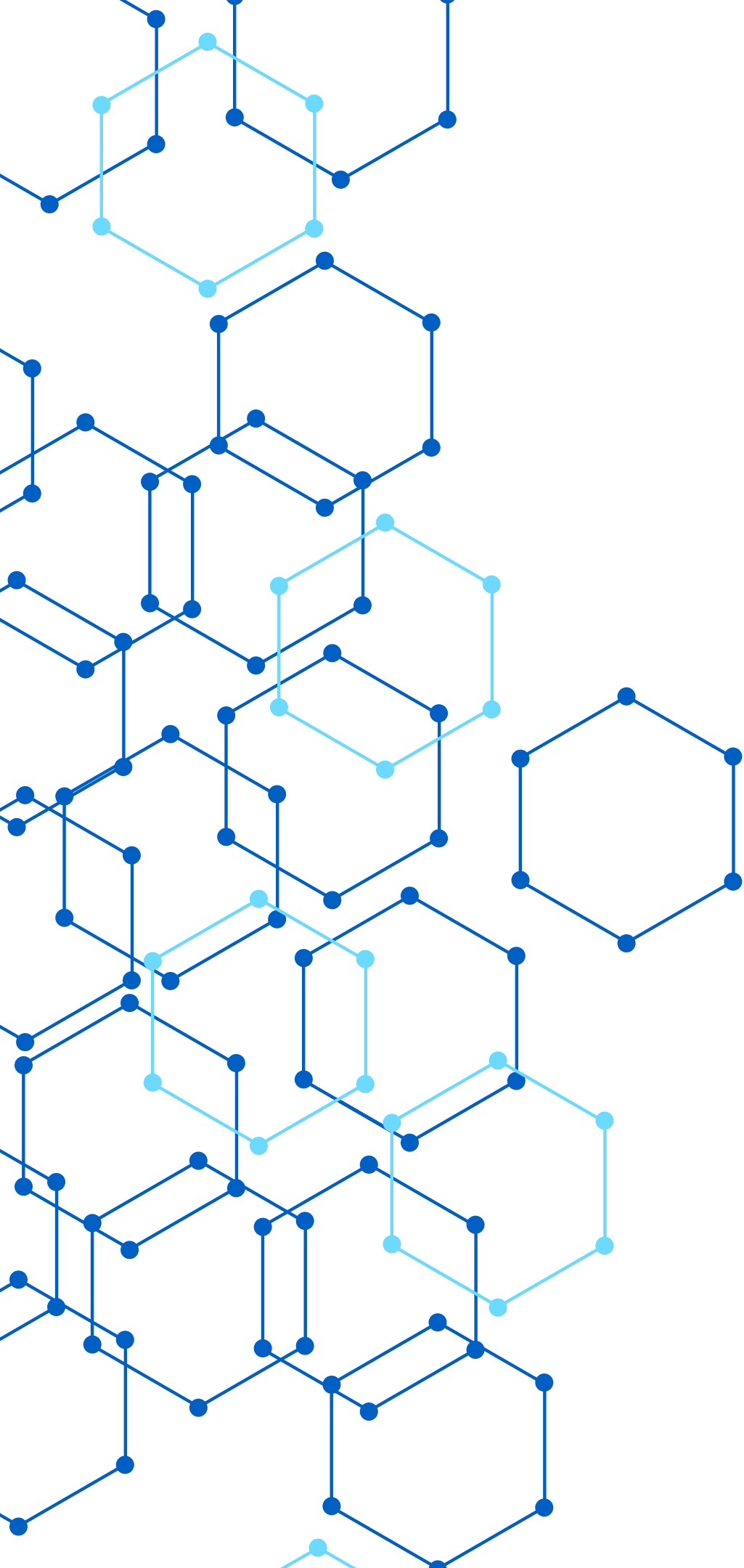
26.10.1 CREATING A DECISION TREE

A decision tree provides a systematic way to evaluate make/buy alternatives.

For each option:

- Identify possible outcomes
- Assign probabilities
- Estimate the cost of each outcome
- Calculate expected cost
- Compare alternatives





EXPECTED COST

EXPECTED COST = Σ (PATH PROBABILITY $\times$ ESTIMATED PATH COST)

EXAMPLE:
BUILD

- SIMPLE: 30% $\times$ \$380K
- DIFFICULT: 70% $\times$ \$450K

EXPECTED COST = \$429K

EXAMPLE RESULTS

Option	Expected Cost
Build	\$429K
Reuse	\$382K
Buy	\$267K
Contract	\$410K

LOWEST EXPECTED COST: BUY

COST ISN'T EVERYTHING

Other decision criteria include:

- Availability
- Developer experience
- Vendor/contractor experience
- Conformance to requirements
- Local organizational "politics"
- Likelihood of future change

26.10.2 OUTSOURCING

Contracting software engineering activities to a third party

Why outsource?

- Q Potential cost savings
- Q Access to external expertise
- Q Potentially higher quality
- Q Reduced internal staffing/infrastructure

STRATEGIC VS TACTICAL



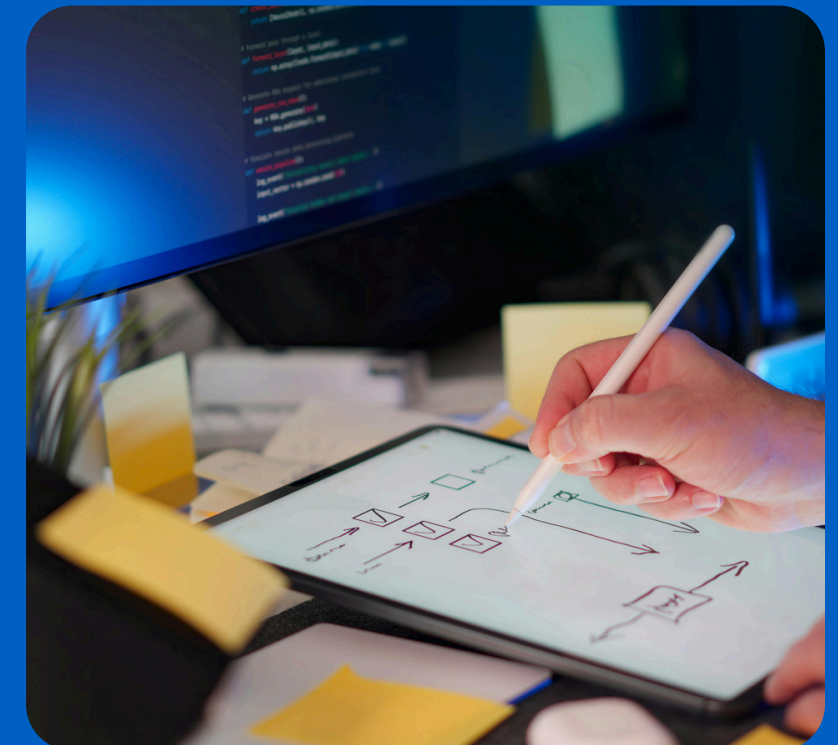
Strategic

- Business managers decide whether a significant portion of software work should be contracted out



Tactical

- Project manager determines whether part or all of a specific project should be outsourced



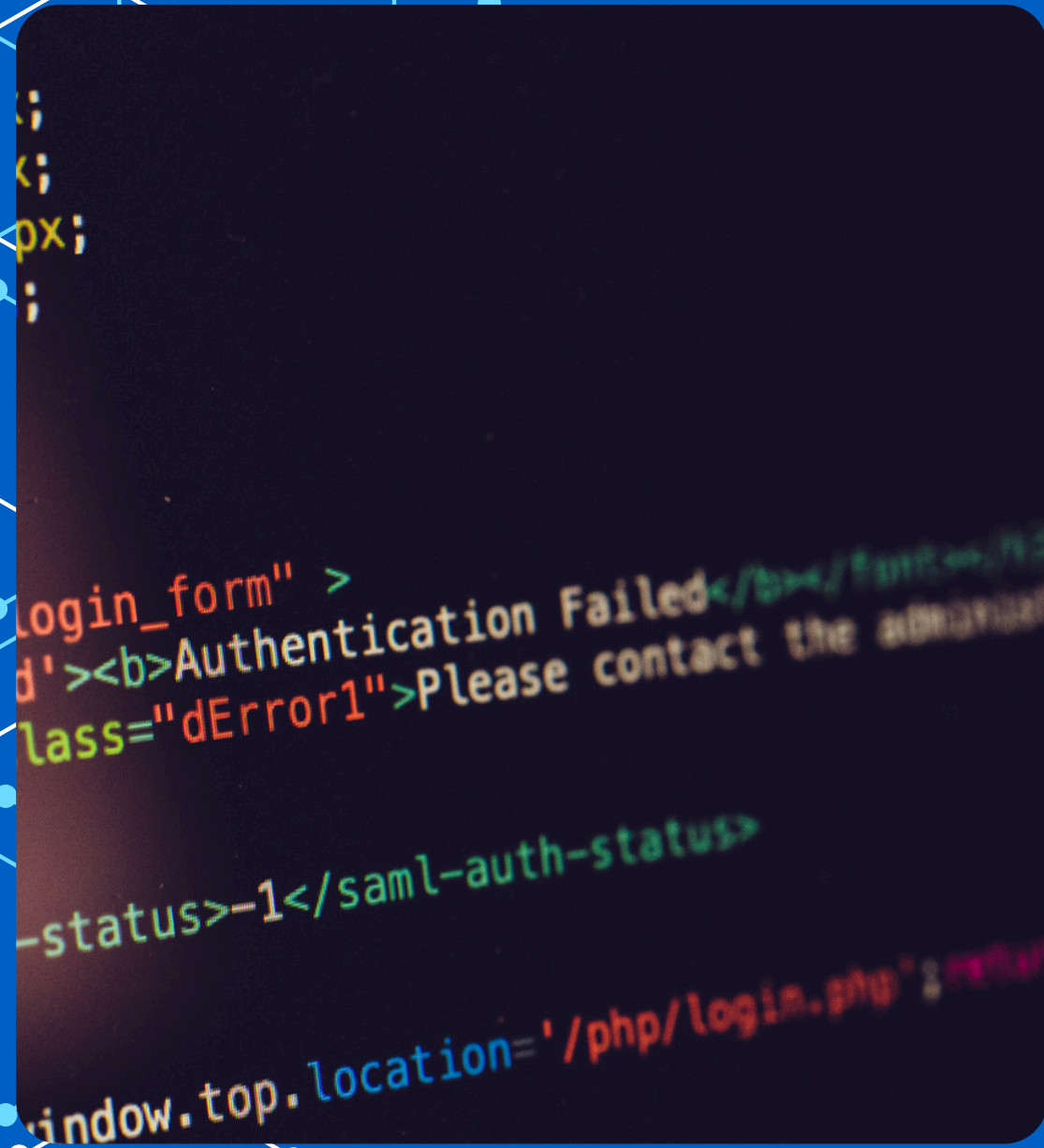
BENEFITS VS. RISKS

Potential Benefit

- Cost savings

Potential Risks

- Loss of control
- Dependence on third party
- Risk to competitiveness
- Quality and coordination concerns
- Need for strong management



The image shows a dark rectangular box with a blurred background. Inside the box, there is a screenshot of a computer screen displaying code and an error message. The visible text includes: `login_form" >`, `<b>Authentication Failed</b></font></td>`, `class="dError1">Please contact the administrator`, `-status>-1</saml-auth-status>`, and `window.top.location='/php/login.php'; return`. The text is in various colors (red, green, blue) typical of a code editor or terminal window.



26.11 SUMMARY

Before a project begins, estimate:

1. How long will it take?
2. How much effort will be required?
3. How many people will be involved?
4. What resources will be needed?
5. What risks are involved?

TWO BROAD ESTIMATION CATEGORIES



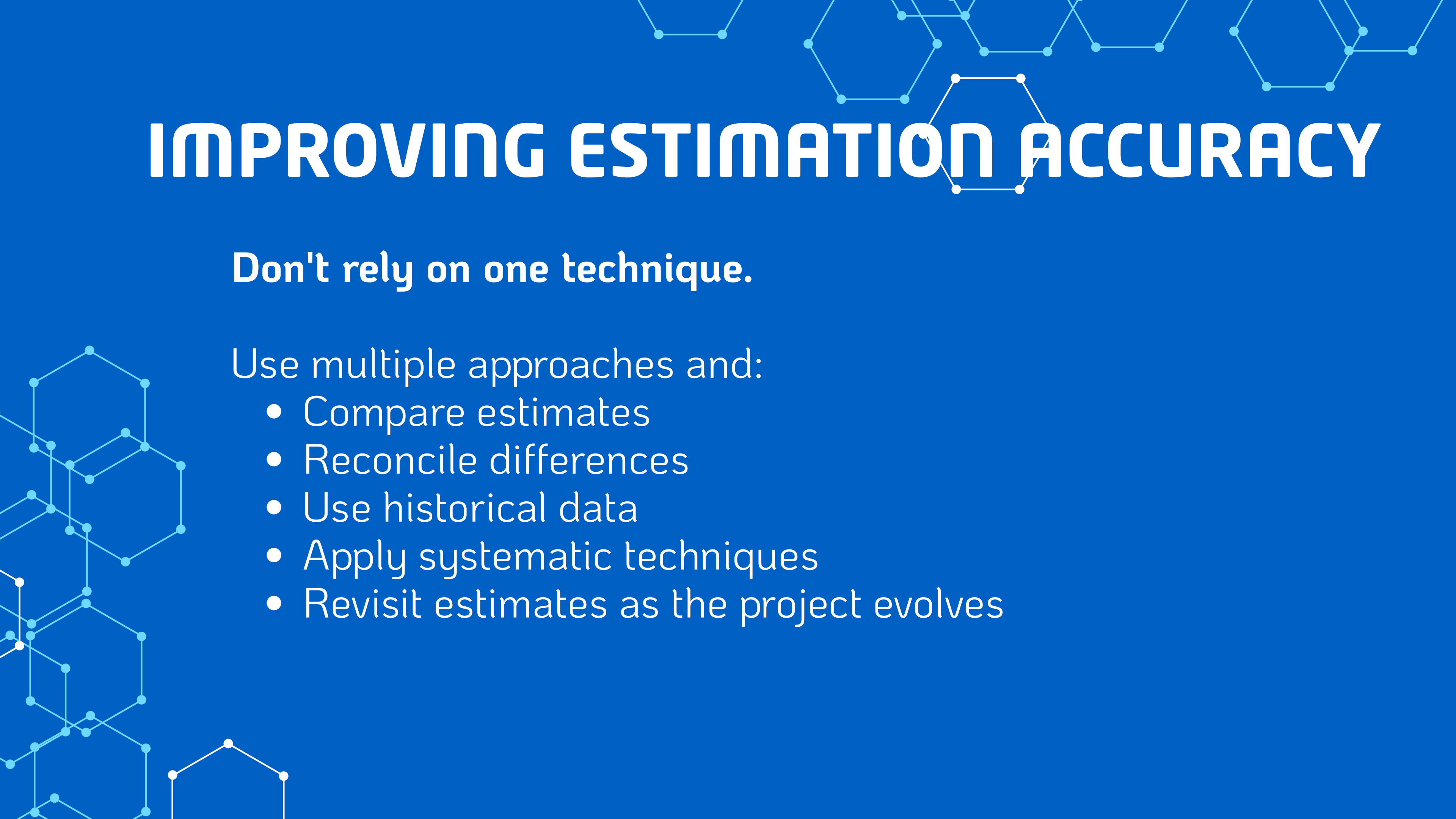
Decomposition

- Break software into functions/tasks
- Estimate LOC, FP, use cases, person-months, etc.



Empirical Modeling

- Use empirically derived formulas
- Predict effort and development time
- Estimation tools can automate models

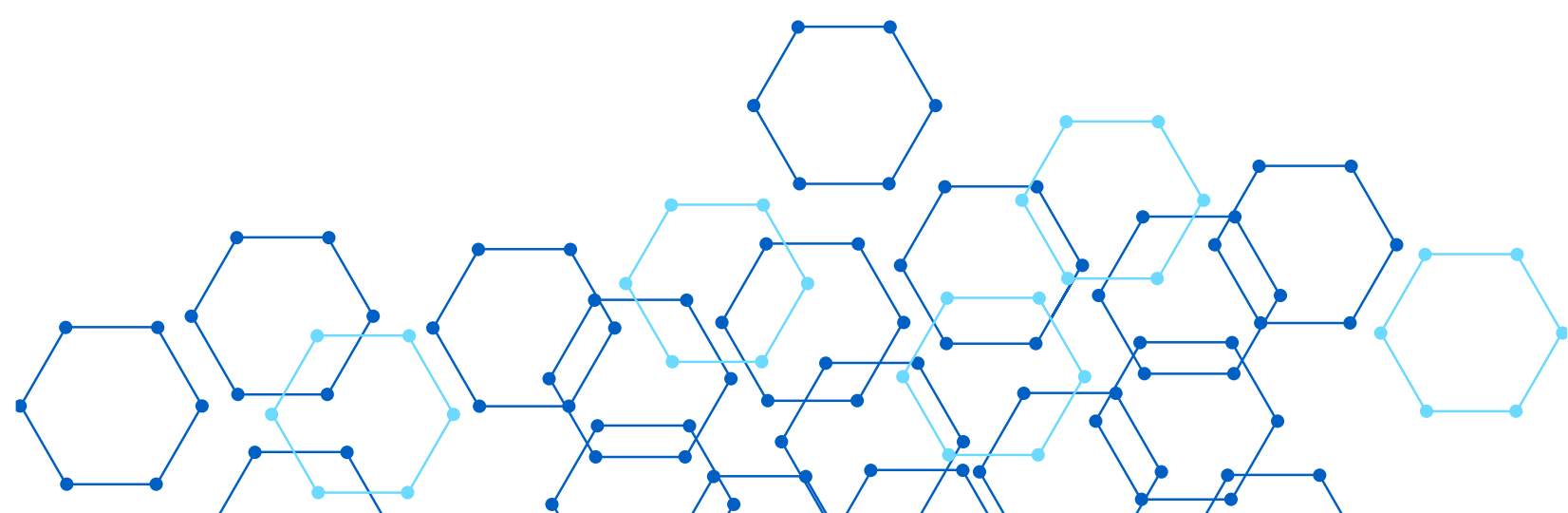
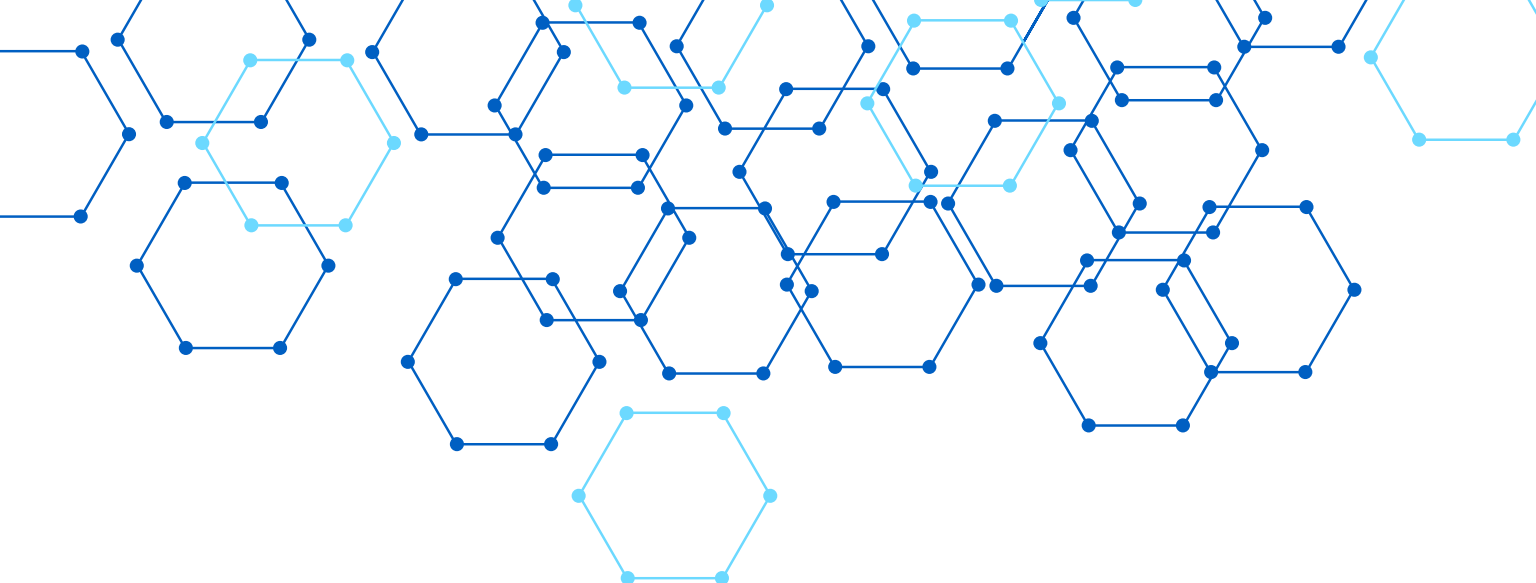
A decorative background pattern consisting of a network of light blue lines and dots, forming various hexagonal and polygonal shapes, primarily concentrated in the top and bottom corners.

IMPROVING ESTIMATION ACCURACY

Don't rely on one technique.

Use multiple approaches and:

- Compare estimates
- Reconcile differences
- Use historical data
- Apply systematic techniques
- Revisit estimates as the project evolves



THANK YOU

Presented by
Añora, John Ray Donovan
Aquino, Ronaldvince Yrlwin
Villareal, Renson Nicolai